

ECE 3210 quiz03

Name: _____ SOLUTIONS _____

1. (10 points) Find $y(t)$ if

$$y(t) = t \cdot (u(t) - u(t-1)) * e^{-t}u(t).$$

Note: this is essentially the same as the lab convolution.

Solution: First, we will need to choose one of the signals to flip. The easier of the two is the $e^{-t}u(t)$ signal. Flipping it gives $e^{-(t-\tau)}u(t-\tau)$. We see that if we shift to the left $t < 0$ then $y(t) = 0$.

Case 1: $0 < t < 1$

We will need to set up the integral for this situation, which is

$$y(t) = \int_0^t \tau e^{-(t-\tau)} d\tau$$

We see we can pull out the e^{-t} term and integrate by parts. This gives

$$y(t) = e^{-t} [\tau e^{\tau} - e^{\tau}]_0^t = e^{-t} [te^t - e^t + 1] = t - 1 + e^{-t}$$

Case 2: $t > 1$

We will need to set up the integral for this situation, which is

$$y(t) = \int_0^1 \tau e^{-(t-\tau)} d\tau$$

We see we can pull out the e^{-t} term and integrate by parts. This gives

$$y(t) = e^{-t} [\tau e^{\tau} - e^{\tau}]_0^1 = e^{-t} [e^1 - e^1 + 1] = e^{-t}$$

Finally, we can combine the two cases into one piecewise function

$$y(t) = \begin{cases} 0 & t < 0 \\ t - 1 + e^{-t} & 0 < t < 1 \\ e^{-t} & t > 1 \end{cases}$$